Ishaan K. Marwaha

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Objective

Computer Engineering student with experience in machine learning for data processing, embedded systems, and computer hardware engineering across scientific research, extracurricular organizations, and self-driven entrepreneurship experiences. Seeking research opportunities and internship positions in computer engineering across Spring 2027 and Summer 2027.

Education

Georgia Institute of Technology | Atlanta, GA *Expected Graduation: December 2028*
Bachelor of Science in Computer Engineering | John H. Martinson Honors Program | GPA 4.00

Skills

Programming Languages: Python, C, C++, Verilog, SystemVerilog, MATLAB, Java
Hardware: PCB Design, Embedded Systems, CPU Design and Verification
Frameworks/Platforms: PyTorch, Pandas, OpenCV, Git/GitHub, Linux, KiCAD

Experience

Bhasin Systems Biomedicine Lab, Emory University | Atlanta, GA *May 2024 – August 2026*
Research Intern / Department of Bioinformatics
Student-run team focusing on developing machine learning models for early-stage cancer biomarker detection based on sparse RNA sequencing data.

- Faced with computational bottlenecks when processing highly sparse single-cell transcriptomic datasets, engineered a memory-efficient machine learning pipeline in PyTorch, achieving high predictive precision against industry-standard tests.
- To validate this methodology for scalable application, benchmarked the algorithm's performance, error rates, and sparse data handling against traditional statistical methods, culminating in a first-author publication in *Frontiers in Aging* (2025).

ViTAL Lab, Georgia Tech / Emory University | Atlanta, GA *July-August 2024*
Research Intern / Department of Biomedical Engineering
Summer research experience developing comprehensive data pipelines to extract joint angles and gait data from videos using ML

- Needing an efficient method to extract spatial data from continuous video recordings, developed a Python-based computer vision and time-series processing pipeline using NumPy and Pandas, enabling accurate, quantitative biomechanical analysis.

Medi-Loc LLC *May 2025 - Present*
Co-Founder & Lead Developer
Co-founded a startup to develop a smart medication dispenser aimed at improving medication adherence. Team-based project.

- To transition an automated medical device from concept to a functional prototype, architected the custom mainboard schematic and production PCB layout, resulting in a reliable low-cost hardware platform capable of precise actuation.
- Tasked with ensuring accurate mechanical control and peripheral communication across the custom board, developed low-level C/C++ embedded firmware, achieving real-time sensor polling, motor timing, and system-level fault handling.

Projects

RISC-V Processor – Georgia Tech Silicon Jackets *August 2026 - Present*
Implemented several core components of the RISC-V CPU architecture as an individual project.

- Aiming to deepen expertise in silicon design, engineered essential datapath and control unit modules as well as basic functionality, successfully translating Instruction Set Architecture (ISA) specifications into functional logic.
- To verify instruction execution and system reliability, developed comprehensive testing protocols including a complete constrained random verification pipeline.

TurboSentry Jet Engine Monitoring Pipeline *January 2026 - June 2026*
Developed a GPU-accelerated deep learning pipeline to process jet engine telemetry and predict remaining life in real time.

- To feed complex multivariate sensor streams into deep temporal models, designed an end-to-end ingestion pipeline that serialized raw NASA telemetry over multithreaded TCP sockets and validated outputs with industry-standard tests.
- In order to track nonlinear degradation patterns across assets, developed a Deep Bidirectional LSTM network in PyTorch, resulting in a highly accurate model with an average failure prediction error of under 1 operational cycle.

Publications

Bhasin, A., Marwaha, I.K.*, & Nooka, L.S.* (2025). Estimation of biological aging based on T-cell differentiation trajectories: emerging and future avenues. Frontiers in Aging, 6. [* = Shared First Authorship]*

Relevant Coursework

Intro to Computing (CS 1301); Linear Algebra; Multivariable Calculus; Differential Equations; Digital System Design (Spring 2027)